

■ Scientific Justification (required for all)

Does an AGN’s infrared spectrum remember its accretion history? An active galactic nucleus (AGN) can fade while its warm dust still radiates in response to an earlier bright state; after brightening, the dust can lag behind the new nuclear power. Strongly variable quasars [1,2], including changing-look AGN [3], let us measure this mismatch against a recorded luminosity history. The unresolved issue is the amplitude and persistence of this warm spectral imprint, and whether delayed heating alone explains it.

We request **67.0 h of MIRI/MRS spectroscopy of 24 AGN at $z < 0.3$: 12 fading and 12 rising sources**. Every target has ZTF/WISE light curves, SPHEREx spectra and multi-epoch optical spectroscopy, including 2026 Palomar/NGPS spectra. These records constrain the nuclear history and hot-dust response. One MIRI epoch per target adds the warm continuum, both silicate features and spatial constraints on host emission. The programme addresses two linked questions:

1. How strongly does the warm infrared spectrum retain recent accretion history? We will measure the warm-continuum excess or deficit relative to equilibrium at the current nuclear power, and its dependence on the direction, amplitude and age of the luminosity change.

2. Does this memory require a change in the dust distribution? Light-travel delays imprint history even on a fixed distribution. We will test whether this response explains the spectra or whether the amount or location of the emitting dust must also change.

From infrared echoes to dust evolution. WISE observations have established infrared echoes of changing-look transitions [4–7]. Broadband colours, however, combine temperature, emitting area, extinction and host light [8]. The proposed spectra test the warm spectral signature of those histories and constrain which properties of the emitting material produce it.

The equilibrium sublimation radius scales approximately as $L^{1/2}$ [9–11], but the physical dust boundary can adjust over years [12]. Individual nuclei already suggest different responses: the smaller dust radius after Mrk 590 faded supports replenishment [13], whereas the temperature variations in Mrk 817 favour heating and cooling at a nearly fixed boundary [14]. These results motivate a comparison across both directions of change. Recovering sources further test whether the spectrum depends on the path to its present state.

MIRI’s diagnostic power. SPHEREx covers 0.75–5 μm [15], constraining hot dust close to the nucleus. Longer delays and weaker variability at longer wavelengths [16,17] make the warm continuum sensitive to earlier illumination. MRS supplies the missing 4.9–27.9 μm spectrum [18]: the continuum measures the imprint, the paired silicate features constrain geometry and optical depth [19], and spatial profiles and aromatic bands constrain host emission [20]. These diagnostics are needed together to interpret a warm excess or deficit. The result will measure, or bound, the bias from recent variability when equilibrium infrared templates are used to infer nuclear power and dust covering factor in these strongly variable AGN.

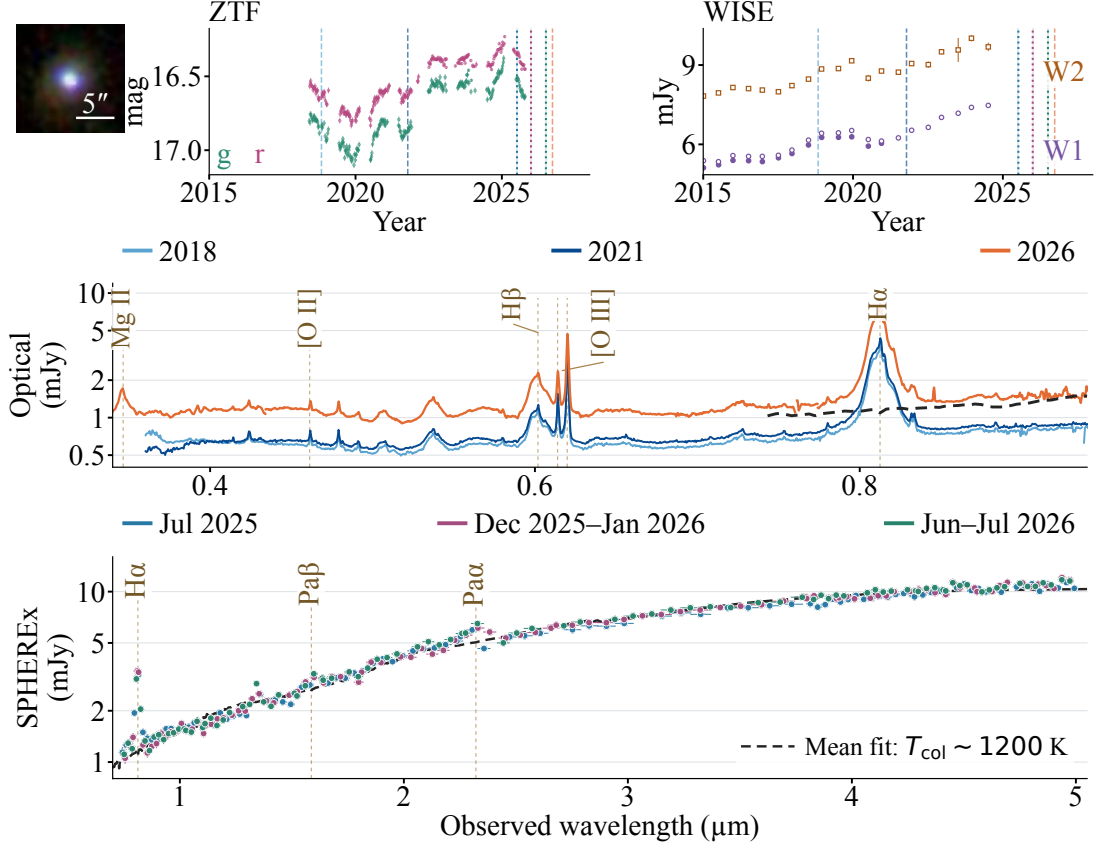


Figure 1: **Connecting nuclear variability to the hot-dust spectrum.** P9694 ($z = 0.238$): SDSS image (5'' bar), ZTF/WISE light curves, optical spectra from 2018, 2021 and 2026, and three SPHEREx visits. Dashed curves show the same pooled host+disc+blackbody fit over the measured wavelength overlap ($T_{\text{col}} \sim 1200$ K, model-dependent). This mean fit does not measure temperature evolution. Colours link spectra to light-curve dates; shading spans SPHEREx visits. Wavelengths are observed-frame and spectral fluxes are in mJy on logarithmic axes, without rescaling. Gold markers indicate expected lines.

A sample selected by variability history. The sample combines catalogued changing-look AGN [21–24] and large-amplitude variables selected by ZTF/WISE history, primarily through a low-dimensional variability map. P9694 illustrates a rising source (Figure 1). We follow full histories, including P11530’s fading in 2018–2021 and partial recovery in 2022–2023. The 2026 NGPS spectra extend archival continuum and broad-line constraints.

History	Targets	Redshift range	Archival W3 (mJy)
Fading	12	0.014–0.269	2.8–48.4
Rising	12	0.017–0.295	1.9–32.6

W3 fluxes include host light; the science comparison uses decomposed nuclear luminosities.

Measuring the spectral memory of accretion. Our primary observable is the logarithmic ratio of nuclear 8–13 μm continuum luminosity to its equilibrium prediction at current nuclear power (Figure 2). The balanced rising and fading groups provide the comparison: we will fit their offset jointly with luminosity, change amplitude and event age, allowing object-specific geometry, covering factor, extinction and host emission. Comparisons within the overlapping luminosity range and with individual targets omitted will establish how consistently the offset is supported across the sample.

Placing each spectrum in its response history draws on all three existing data sets. Optical decomposition supplies the nuclear history, WISE constrains response amplitudes and measurable delays, and SPHEREx constrains hot dust at its observed epochs. Rest-frame event ages are compared with measured lags or radius–luminosity priors [11,17,25], with warm delays inferred from each geometry. Uncertainty in event dates, gaps and the conversion from optical to dust-heating luminosity propagates into each target’s response stage.

The spectra then constrain the interpretation through silicate profiles, spatial structure and aromatic bands [19,20]. Equilibrium and history-dependent models will share geometry and host priors. Mock MRS spectra spanning the allowed histories, nuclear fractions and geometries, with statistical and calibration covariance included, will pass through the same analysis to calibrate false-positive rates, detectable offsets and upper limits.

With twelve sources in each group, the mean-offset uncertainty is $0.408s$ for independent residual scatter s . For planning, $s = 0.20$ dex gives 0.082 dex precision and 80% power to detect a 0.24 dex offset (two-sided t -test, 5% significance). This conditional Gaussian estimate sets a sensitivity scale; the recovery tests extend it to the fitted covariates, correlated errors and individual sample uncertainties.

Testing for dust evolution. We will use clumpy and disc-plus-wind geometries [26–28] to construct responses to the reconstructed histories, allowing for composition and grain-size uncertainty [29]. Fixed distributions respond through grain heating and cooling; evolving distributions also change the amount or location of dust [30]. Both include light-travel delays. The diagnostic

$$\delta_{\text{warm}} = \log_{10} \left(\frac{L_{8-13\mu\text{m}}^{\text{MRS}}}{L_{8-13\mu\text{m}}^{\text{fixed}}(t_{\text{MIRI}})} \right),$$

measures the warm-continuum residual relative to a *delayed fixed-distribution* prediction. The known limitations of static spectral models [20] motivate a joint test: evolving models must explain coherent continuum and silicate residuals across the allowed fixed models while also fitting the earlier data, with host and calibration uncertainties propagated throughout [31]. The hot-to-warm luminosity ratio provides a complementary test where the history constrains the 2–5 μm emission at the MIRI epoch.

Figure 3 illustrates the spectral effect of boundary evolution in optically thin shells. Both models include delayed heating: one retains the dust distribution, while the other moves its inner boundary immediately when changed illumination arrives, an assumed limiting case. Where both hypotheses fit, the primary comparison still measures or bounds the warm spectral imprint of history.

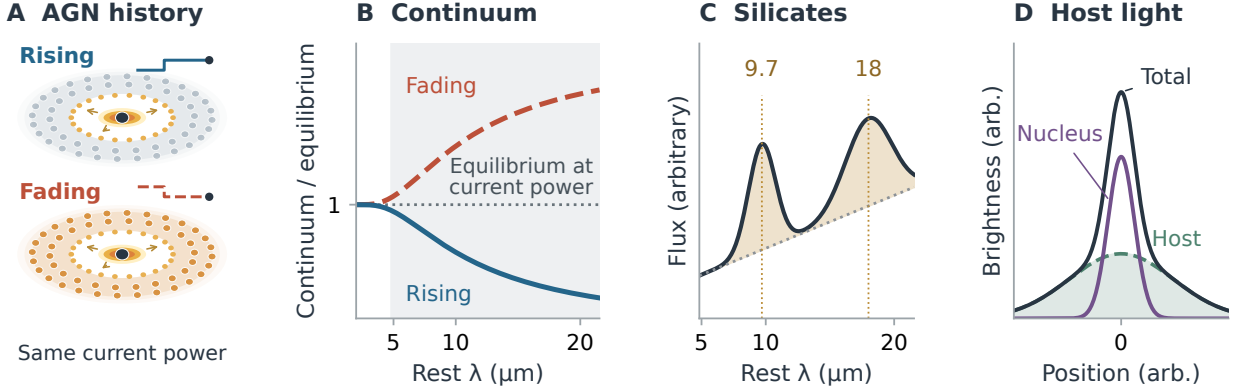


Figure 2: **Question 1: the spectral imprint of past activity.** A: equal current power, different histories. B: schematic delayed responses relative to current-power equilibrium (unity); shading marks common MRS coverage. C–D: silicates and spatial profiles constrain geometry, obscuration and host light [19,20]. All panels are schematic.

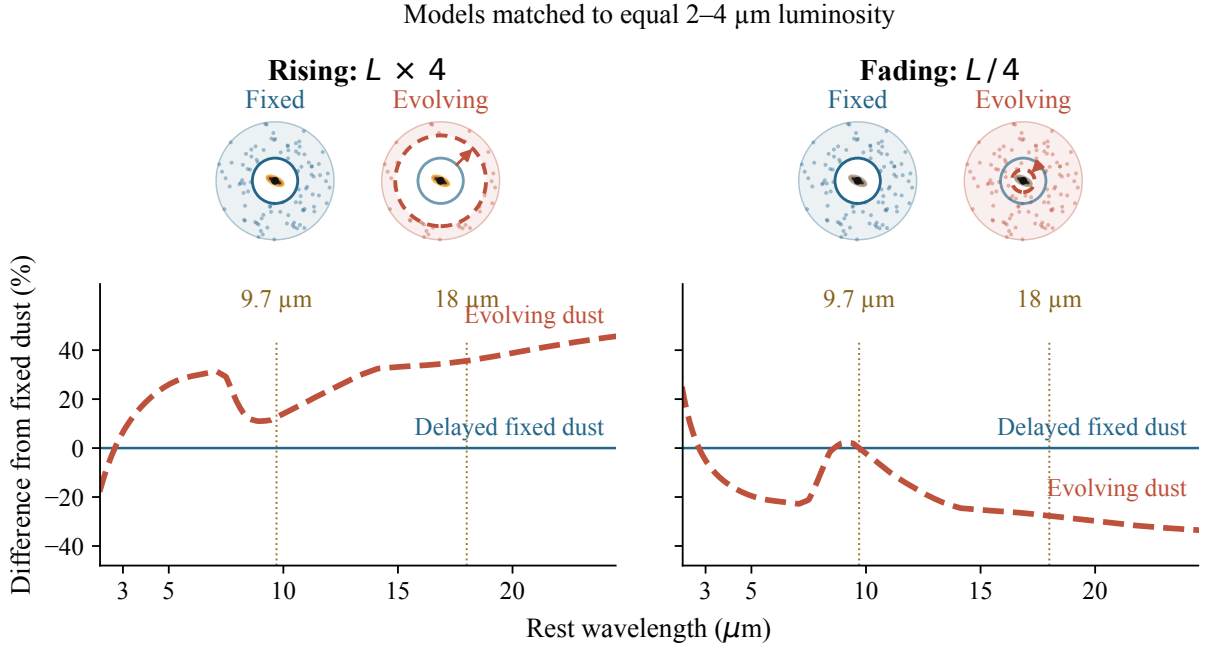


Figure 3: **Question 2: is dust evolution required?** Illustrative optically thin shells use Laor & Draine's silicate/graphite optical properties [32]. Both models share a fourfold rise/decline and light-travel delays; the evolving boundary follows $R_{\text{in}} \propto L^{1/2}$. Spectra at $t = 20R_{\text{in,Si},0}/c$, where $R_{\text{in,Si},0}$ is the initial silicate inner radius, are normalised to equal 2–4 μm luminosity. The ordinate is $100(F_{\text{v}}^{\text{evolving}}/F_{\text{v}}^{\text{fixed}} - 1)$; positive means brighter with evolution. Shells illustrate boundary physics; recovery tests with the full models will establish detectability.

■ Description of Observations (required for all GO and Survey)

Spectral coverage and sensitivity. We request one MIRI/MRS science visit and a matching sky visit per target. All three grating settings, nuclear target acquisition and a four-point dither provide approximately 4.9–27.9 μm coverage [18,33]. Both silicate peaks fall within this range for all 24 targets. The primary rest-frame 8–13 μm continuum diagnostic is accessible throughout the sample without channel 4C; the full spectrum constrains the longer-wavelength continuum and 18 μm profile.

Archival total-source W3 and W4 fluxes span 1.9–48.4 and 6.4–82.6 mJy. We target continuum S/N of 50 per $R = 100$ bin near rest 5 and 12 μm and 30 around the 18 μm feature, corresponding to 2% and 3.3% statistical errors per bin. We use FULL/FASTR1, one integration per dither, and 27/60/27 groups in settings A/B/C for 23 targets. Four dithers give 299.7/666.0/299.7 s on source, with all four channels observed simultaneously. The W3-faint J083826.50+371906.7 uses 80 groups in B (888.0 s). Equal-depth sky exposures preserve extended host emission for decomposition [33]. Science-plus-sky integration totals 17.00 h. **We request 67.0 h**, including acquisition, visit operations, slews and observatory overheads calculated with APT 2026.5.1: a Medium programme [34].

For J083826.50+371906.7 at $z = 0.2111$, ETC/Pandeia 2026.7 predicts S/N of 55/225/61 per $R = 100$ bin at rest 5/12/18 μm . These estimates assume a point source with $F_{\nu} \propto \lambda^2$, 0.95 mJy in W3 (half the archival total), the adopted sky spectrum, equal-depth sky exposures and extraction radii of 0.3"/0.6"/0.9". They establish statistical sensitivity under the adopted nuclear-flux model; host and calibration uncertainties enter the science inference separately. Acquisition uses F560W for the faintest sources and the neutral-density filter for brighter nuclei, with readouts giving S/N above 50 without saturation over the adopted flux ranges [35].

Connecting the observing epochs. Fitting WISE and SPHEREx at their actual dates places each MIRI spectrum within its illumination history. Optical monitoring near JWST anchors the nuclear continuum to the MIRI epoch; spectroscopy constrains spectral-shape and broad-line changes. Joint host and aperture modelling connects the epochs. Histories consistent with the photometry bracket gaps and propagate current-power uncertainty into the warm residual. Repeating the comparison for targets with well-constrained MIRI-epoch luminosities tests its stability. Together, these records allow one MIRI epoch per target to sample the ensemble response.

Separating nuclear dust from the host. We will fit each MRS cube with an unresolved nucleus and extended emission [20], then model the continuum, silicates, aromatic bands, H_2 and ionic lines [36]. Alternative point-spread-function extractions and host templates will define the uncertainty in the nuclear continuum. Aromatic bands constrain star formation within the unresolved component; [Ne V] and [Ne VI] trace highly ionised gas. Ambiguous nuclear decompositions will enter as bounds rather than precise fluxes.

For the 2010 W3/W4 constraints, we will integrate redshifted nuclear-plus-host models through the WISE bandpasses, accounting for colour corrections and emission outside MRS coverage. Archival imaging constrains host light beyond the MRS field and within the SPHEREx aperture. Nuclear-flux bounds and wavelength-dependent calibration covariance [31] propagate through both science tests.

Supplemental Information

Special requirements. Each science/sky pair is linked as a consecutive, noninterruptible sequence to obtain a contemporaneous background [33]. Sky fields are selected using WISE and 2MASS catalogues and images, with at least $25''$ clearance from catalogued sources. Twenty-three fields lie within 1–2 arcmin of their nuclei; the crowded field around J20145928+2523010 uses a $135''$ offset. Background targets are marked extended and use four-point extended-source dithers, with no target acquisition. No fixed-date constraints, coordinated parallels or joint-facility time are requested. APT’s Visit Planner returns scheduling windows for all 48 visits.

Duplications. A coordinate-based MAST search on 2026 September 26, including archived and planned JWST observations, found no overlapping observations within $120''$ of any of the 24 science positions. No potential MIRI/MRS duplications were identified.

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